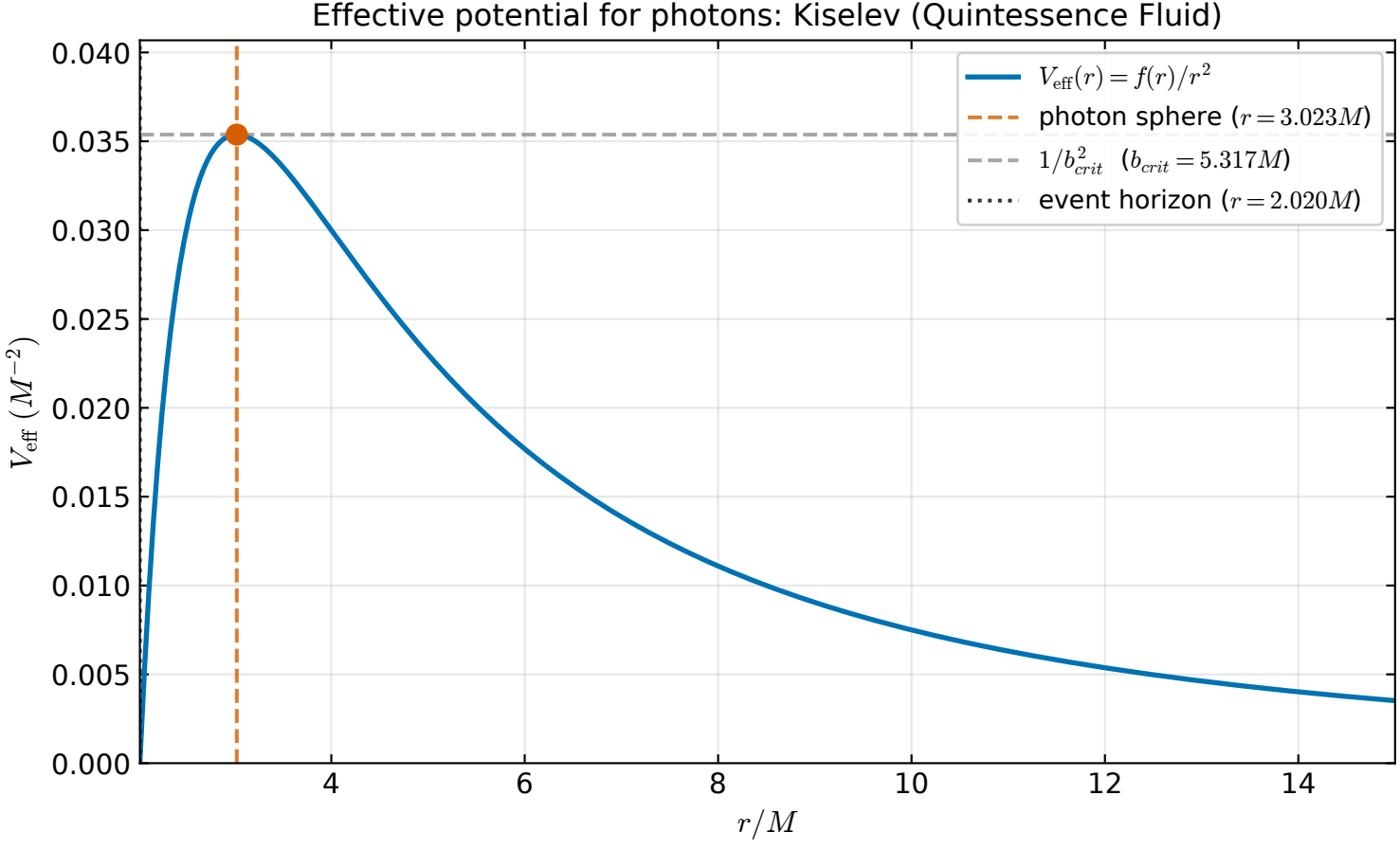
Effective potential for photons: Kiselev (Quintessence Fluid)
$V_{\rm eff}(r) = f(r)/r^2$
photon sphere ($r = 3.023M$)
$1/b_{crit}^2$ ($b_{crit} = 5.317M$)
event horizon ($r = 2.020M$)
$V_{\rm eff}\ (M^{-2})$
r/M
0.000
0.005
0.010
0.015
0.020
0.025
0.030
0.035
0.040
4
6
8
10
12
14